

Small Rotation Blocks and the Order-Six Stable Marriage Problem

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Abstract

We prove that a finite poset with at most fifteen elements, width at most three, and the following common-incomparability property has at most 48 antichains: for every $x < y$, the elements incomparable with both x and y form a chain of size at most two. An explicit fifteen-element poset attains the bound. We show that order-six stable-marriage rotation posets satisfy these hypotheses by examining convex rotation blocks and their fixed partner pools. This gives a structural upper bound of 48 stable matchings, matching the lower bound already provided by Benjamin, Converse, and Krieger in 1995. Equality forces fifteen rotations, all of size two, and makes all 36 man–woman pairs attainable. Accompanying Lean files verify the fixed-token, abstract support, and finite-poset components; the connection to classical rotation semantics is given in the human proof.

1 Introduction

For a strict complete stable-marriage instance I , let $\mathcal{S}(I)$ denote its set of stable matchings and put

$$f(n) = \max_I |\mathcal{S}(I)|,$$

where the maximum is over instances with n men and n women. Determining this maximum is a classical problem; see Gusfield and Irving [5]. Benjamin, Converse, and Krieger [1] constructed an order-six instance with 48 stable matchings in 1995. Thurber [8] subsequently developed further lower-bound constructions. Here we prove the matching order-six upper bound by a local restriction on rotation posets and a sharp abstract poset theorem.

The use of rotation posets and their downsets is classical [2]. Karlin, Oveis Gharan, and Weber [6] obtain an exponential bound by extracting structural properties of these posets, and Palmer and Pálvölgyi [4] give a substantially smaller exponential bound using an entropy argument. We work within this established framework. In particular, the downset correspondence, shared-agent comparability, and the global rotation bound are not new ingredients of the present argument.

Our local input concerns an order-convex block Q of rotations. If the block uses b men, a consecutive execution keeps their partners in a fixed pool and gives $2|Q| \leq b(b-1)$. For three men, a token-return obstruction sharpens the conclusion to $|Q| \leq 2$. This differs in its hypotheses and parameters from the support estimate in [6, Claim 4.7], which applies to arbitrary rotation subsets and counts agents of both sexes. For comparable rotations $\alpha < \beta$, our argument places the rotations incomparable with both on at most $N-3$ men in an order- N instance. At $N=6$, this yields the condition called CI2 below.

The principal counting result is that every poset with at most fifteen elements, width at most three, and CI2 has at most 48 antichains. Its proof couples the counts of two- and three-element antichains, rather than classifying the possible posets. We give an explicit extremal poset and record necessary equality conditions for stable-marriage instances. The resulting local-to-global upper-bound proof does not use an exhaustive classification of preference profiles or saturated rotation

instances. The consecutiveness property of an already ordered family of maximal antichains is known [3]; the relevant step here is that our hypotheses order the entire family of three-element antichains and allow the two counts to be coupled.

A 2026 departmental project announcement at Biola University reports the same numerical result; at the time of writing, we were unable to locate a public proof, and we make no priority claim.¹

2 Rotation blocks

We use rotations and exposure in their standard sense [2, 5]; for the definition and the elimination operation, see also [6, Definitions 4.1 and 4.4]. We recall their properties, rather than redefine a rotation by its list representation. An exposed rotation is represented by a cyclic list $((m_0, w_0), \dots, (m_{k-1}, w_{k-1}))$ of matched pairs satisfying the standard rotation conditions, with $k \geq 2$ distinct participating men and distinct current partners. An arbitrary list of matched pairs is not a rotation. Elimination replaces (m_i, w_i) by (m_i, w_{i+1}) , with indices modulo k , leaves non-participants unchanged, and yields a stable matching. Each participating man strictly prefers w_i to w_{i+1} .

The Irving–Leather correspondence [2] identifies stable matchings with downsets of the rotation poset. A downset is realized by eliminating its rotations from the man-optimal stable matching in a linear extension of the precedence order; see [5] and [6, Theorem 4.5]. Rotations sharing an agent are comparable [6, Claim 4.6].

For a rotation ρ , write $\text{supp}(\rho)$ for its set of participating men. For Q a set of rotations, put

$$B(Q) = \bigcup_{\rho \in Q} \text{supp}(\rho).$$

Lemma 2.1 (Convex rotation windows). *Let Q be an order-convex subset of the rotation poset. Then there are stable matchings M_0, M_1 such that, starting from M_0 , the rotations of Q can be eliminated consecutively in a linear extension of the induced order on Q , producing M_1 .*

Proof. Let

$$I = \downarrow Q, \quad J = I \setminus Q.$$

The set I is a downset. We claim that J is also a downset. Let $y \in J$ and $x \leq y$. Since $y \in I$, there is $z \in Q$ with $y \leq z$, and hence $x \in I$. If $x \in Q$, then

$$x \leq y \leq z, \quad x, z \in Q,$$

so convexity would imply $y \in Q$, a contradiction. Hence $x \in J$.

Choose a linear extension of J , followed by a linear extension of the induced order on Q . This concatenation is a linear extension of I : every predecessor of an element of J again belongs to J , while a predecessor of an element of Q belongs to J or to Q . By the standard realization theorem for closed sets of rotations, this gives a valid elimination sequence for I . Its J -prefix reaches the stable matching corresponding to J , and its remaining suffix eliminates exactly the rotations in Q . $\square$

¹Biola University, Department of Math and Computer Science, “Current Students: Completed Projects,” entry for Roman Parks (2026), <https://www.biola.edu/math-and-computer-science-department/current-students>, accessed September 11, 2026.

Lemma 2.2 (Small-block capacity). *Let Q be a finite order-convex set of rotations and put $b = |B(Q)|$. Then every man in $B(Q)$ occurs in at most $b - 1$ rotations of Q . Hence*

$$2|Q| \leq b(b - 1).$$

Moreover,

$$b \leq 2 \implies |Q| \leq 1, \quad b \leq 3 \implies |Q| \leq 2.$$

Proof. Execute Q as the consecutive window supplied by Lemma 2.1. Let W be the set of women matched to $B(Q)$ at the beginning of this window. Throughout the window, the men of $B(Q)$ remain matched bijectively to exactly the same set W . Indeed, at each step the current partners of the participating subset are merely cyclically permuted, while every other man of $B(Q)$ is unchanged.

Fix $m \in B(Q)$. Whenever m participates, his partner becomes strictly worse; between such steps his partner is unchanged. Thus the initial partner of m and the partners immediately following his successive participations are all distinct elements of W . Since $|W| = b$, the man m participates at most $b - 1$ times. Summing over men gives

$$\sum_{\rho \in Q} |\text{supp}(\rho)| \leq b(b - 1).$$

Every rotation has at least two participating men, proving $2|Q| \leq b(b - 1)$. The assertion for $b \leq 2$ follows immediately.

Suppose now that $b \leq 3$ and $|Q| \geq 3$, and consider the first three rotations in the window. They contribute at least six man–rotation incidences. On the other hand, each of at most three men can occur at most twice. Equality is therefore forced: there are exactly three men, all three rotations have support two, and every man occurs in exactly two of the three supports. Thus, naming x as the man common to the first and third supports and a, b as the other two men, the supports in temporal order are

$$\{x, a\}, \quad \{a, b\}, \quad \{x, b\}$$

in that order.

A rotation with exactly two participating men swaps their current partners: all nonparticipants are fixed, and the two current partner tokens must be exchanged. Consequently the three support-two eliminations above return the man common to the first and third supports to his initial partner. That man participated twice, however, and became strictly worse at each participation. He cannot return to his initial partner. This contradiction proves $|Q| \leq 2$. $\square$

Remark 2.3 (Restriction to the participating agents). There is a complementary interpretation of a nonempty convex block Q . Let $J = (\downarrow Q) \setminus Q$, let $B = B(Q)$, and let W be the partners of B in the matching corresponding to J . For every downset K of the induced poset Q , the set $J \cup K$ is a downset of the full rotation poset. Indeed, if $y \in J$ and $x \leq y$, then $x \in J$ because J is a downset. If $y \in K \subseteq Q$ and $x \leq y$, then $x \in \downarrow Q = J \cup Q$. If $x \in Q$, the induced-downset property of K gives $x \in K$; otherwise $x \in J$. Its stable matching restricts to a stable matching of the instance obtained by restricting preferences to B and W : any blocking pair in the restriction would also block the global matching. If $K_1 \neq K_2$, then $J \cup K_1 \neq J \cup K_2$, so the stable-matching/downset bijection gives distinct global stable matchings. During the Q -window all matching changes occur inside $B \times W$; outside $B \times W$ the matching is fixed. Hence the two global matchings can differ only through their restrictions to $B \times W$, and those restrictions are distinct. Writing $\mathcal{J}(Q)$ for the downsets of Q , we obtain

$$|Q| + 1 \leq |\mathcal{J}(Q)| \leq f(|B|).$$

The first inequality follows from the prefixes of a linear extension of Q . Using the elementary exact values $f(2) = 2$ and $f(3) = 3$, this also recovers the two numerical small-block capacity conclusions. We retain the direct token proof because it supplies the participation/incidence bound used later and does not use those external small-order values.

Remark 2.4. For the full rotation set of an order- N instance, Lemma 2.2 recovers the classical bound $|R| \leq N(N-1)/2$. The point needed here is the local version: the same incidence argument applies to every convex rotation block, with the number of men actually used by that block in place of N . The global bound itself is classical; see, for example, [7].

Corollary 2.5. *An order-six stable-marriage instance has at most fifteen rotations.*

Proof. Apply Lemma 2.2 to the entire rotation set R , which is itself convex. If $b = |B(R)| \leq 6$, then the proof of that lemma gives

$$2|R| \leq b(b-1) \leq 6 \cdot 5 = 30.$$

Hence $|R| \leq 15$. □

Corollary 2.6. *The rotation poset of an order-six instance has width at most three.*

Proof. Rotations in an antichain cannot share a man, since rotations sharing an agent are comparable. Their male supports are therefore pairwise disjoint, and each support has size at least two. Six men can support at most three such rotations. □

3 A local structure theorem for order-six rotation posets

For an element x of a poset, let

$$\text{Inc}(x) = \{z : z \parallel x\}.$$

Proposition 3.1 (Common-incomparability support budget). *Let R be the rotation poset of an order- N stable-marriage instance, and let $\alpha < \beta$. Put*

$$Z = \text{Inc}(\alpha) \cap \text{Inc}(\beta).$$

Then

$$|B(Z)| \leq N-3, \quad \text{width}(Z) \leq \left\lfloor \frac{N-3}{2} \right\rfloor, \quad |Z| \leq \binom{N-3}{2}.$$

Proof. Let

$$A = [\alpha, \beta].$$

The sets A and Z are convex. Convexity of A is immediate. If $z_1 \leq y \leq z_2$ with $z_1, z_2 \in Z$, then $y \leq \alpha$ would imply $z_1 \leq \alpha$, while $\alpha \leq y$ would imply $\alpha \leq z_2$. Thus $y \parallel \alpha$, and the same argument with β gives $y \parallel \beta$.

Every $a \in A$ is incomparable with every $z \in Z$: $a \leq z$ would imply $\alpha \leq z$, while $z \leq a$ would imply $z \leq \beta$. Rotations sharing a man are comparable, so

$$B(A) \cap B(Z) = \emptyset. \tag{3.1}$$

The interval A contains the two distinct rotations α, β . By the two-man part of Lemma 2.2, it must therefore use at least three men. Equation (3.1) gives

$$|B(Z)| \leq N-3. \tag{3.2}$$

An antichain in Z consists of rotations with pairwise disjoint male supports, each of size at least two. Hence

$$\text{width}(Z) \leq \left\lfloor \frac{|B(Z)|}{2} \right\rfloor \leq \left\lfloor \frac{N-3}{2} \right\rfloor.$$

Finally Z is convex, so Lemma 2.2 gives

$$2|Z| \leq |B(Z)|(|B(Z)| - 1),$$

which together with (3.2) yields the stated cardinality bound. $\square$

Theorem 3.2 (Common incomparability). *Let R be the rotation poset of an order-six stable-marriage instance. For every $\alpha < \beta$ in R , the set*

$$Z(\alpha, \beta) = \text{Inc}(\alpha) \cap \text{Inc}(\beta)$$

is a chain and has cardinality at most two.

Proof. Proposition 3.1 gives $|B(Z)| \leq 3$ and $\text{width}(Z) \leq 1$, so Z is a chain. Since Z is convex, the three-man part of Lemma 2.2 gives $|Z| \leq 2$. $\square$

This explains why order six is the natural endpoint of the present local argument. For $Z = \text{Inc}(\alpha) \cap \text{Inc}(\beta)$, Proposition 3.1 gives $|B(Z)| \leq N-3$, $\text{width}(Z) \leq \lfloor (N-3)/2 \rfloor$, and $|Z| \leq \binom{N-3}{2}$. At $N=6$, the bounds $|B(Z)| \leq 3$ and $\text{width}(Z) \leq 1$ force a chain on at most three participating men, and the exceptional three-man capacity sharpens the cardinality bound to $|Z| \leq 2$. At $N=7$, the same general budget gives only $|B(Z)| \leq 4$, $\text{width}(Z) \leq 2$, and $|Z| \leq 6$. Thus this mechanism forces the CI2 rigidity used below at order six, but the same budget no longer forces it at order seven.

Theorem 3.2 motivates the following purely order-theoretic condition.

Definition 3.3. A finite poset P has property CI2 if, whenever $x < y$, the common incomparability set

$$\text{Inc}(x) \cap \text{Inc}(y)$$

is a chain of cardinality at most two.

Proposition 3.4 (Forbidden-pattern form). *A finite poset satisfies CI2 if and only if it has no induced subposet isomorphic to $2+1+1$ or to $2+3$, where the summands are mutually incomparable chains.*

Proof. If CI2 fails for $x < y$, then either $\text{Inc}(x) \cap \text{Inc}(y)$ contains two incomparable elements, which together with $x < y$ induce $2+1+1$, or it contains a three-element chain, which together with $x < y$ induces $2+3$. The converse is immediate from either forbidden configuration. $\square$

Remark 3.5. Equivalently, the incomparability graph of a CI2 poset has no induced diamond and no induced $K_{2,3}$. Thus the pure order-theoretic part of the proof concerns a restricted class of cocomparability graphs. Antichains of the poset are precisely cliques of its incomparability graph.

4 A finite-poset bound

Let P be a finite poset of width at most three, let $n = |P|$, and let A_j denote the number of j -element antichains. Since every antichain has size at most three,

$$|\mathcal{A}(P)| = 1 + n + A_2 + A_3. \quad (4.1)$$

We first record the two bounds needed for the generic case.

Lemma 4.1 (Triple structure). *Assume P has width at most three and satisfies CI2. If $A_3 > 0$, then*

$$A_3 \leq \left\lfloor \frac{n-1}{2} \right\rfloor.$$

Proof. Let T be a three-element antichain and $x \notin T$. The point x is comparable with at least one member of T , else $T \cup \{x\}$ would be a four-element antichain. In fact it is comparable with at least two. Otherwise, if x were comparable with only $t \in T$, the other two members of T would be incomparable elements of $\text{Inc}(x) \cap \text{Inc}(t)$, contradicting CI2. Moreover x cannot lie above one member of T and below another, since those two members would then be comparable. Thus every $x \notin T$ lies below at least two members of T , or above at least two.

Write these alternatives as $x \in L(T)$ and $x \in U(T)$. If $l \in L(T)$ and $u \in U(T)$, the two corresponding two-element subsets of T intersect, so $l < t < u$ for some $t \in T$; hence

$$L(T) < U(T). \quad (4.2)$$

Now let S, T be three-element antichains. We claim that their generated ideals are nested. Suppose first that some $u \in S \setminus T$ lies in $U(T)$. Then no element of $S \setminus T$ can lie in $L(T)$, because (4.2) would make it comparable with u . Thus every element of $S \setminus T$ lies in $U(T)$.

We show that every $t \in T$ lies in $\downarrow S$. This is immediate if $t \in S$. Otherwise suppose that $t \not\leq s$ for every $s \in S$. For $s \in S \cap T$ we have $t \parallel s$. For $s \in S \setminus T$, the point s lies above at least two members of T . Hence $t > s$ is impossible: among those two members choose $t' \neq t$; then $t' < s < t$, contradicting that T is an antichain. Our assumption $t \not\leq s$ therefore forces $t \parallel s$ for every $s \in S$. Then $S \cup \{t\}$ is a four-element antichain, contradicting the width bound. Hence $t \leq s$ for some $s \in S$, and so $\downarrow T \subseteq \downarrow S$.

If no element of $S \setminus T$ lies in $U(T)$, then every such element lies in $L(T)$ and hence below a member of T . Thus $S \subseteq \downarrow T$ and $\downarrow S \subseteq \downarrow T$. This proves nesting.

Distinct three-element antichains generate distinct ideals, since the maximal elements of $\downarrow T$ are exactly the members of the antichain T . Also two distinct triples share at most one point: if $S \neq T$ shared two points, a point of $S \setminus T$ would be incomparable with two members of T , contrary to the first paragraph.

Order all three-element antichains as

$$T_1, \dots, T_t, \quad \downarrow T_1 \subsetneq \dots \subsetneq \downarrow T_t, \quad t = A_3.$$

The following no-re-entry property is the standard consecutiveness property for ordered maximal antichains; see [3, Proposition 2.4 in the preprint version]. It applies here because a three-element antichain is maximal in a width-three poset, and the maximal-antichain order agrees with inclusion of generated ideals. We include the short proof for completeness. If $i < j < k$ and $x \in T_i \cap T_k$, then $x \in T_j$. Indeed $x \in \downarrow T_j$, so $x \leq y$ for some $y \in T_j$; also $y \in \downarrow T_k$, so $y \leq z$ for some $z \in T_k$. Since $x, z \in T_k$ and T_k is an antichain, $x = y = z$.

It follows that each T_{i+1} meets $T_1 \cup \dots \cup T_i$ in at most one point. The first triple contributes three points and every later triple contributes at least two new points. Hence

$$n \geq 3 + 2(t - 1) = 2t + 1.$$

□

Lemma 4.2 (Pair bound). *Under the assumptions of Lemma 4.1, if $A_3 > 0$, then*

$$A_2 \leq 2n - 3.$$

Proof. By Dilworth's theorem, partition P into three chains C_1, C_2, C_3 . Because a three-element antichain exists, each chain is nonempty. Consider one chain

$$C = \{c_1 < \dots < c_a\}.$$

For every $x \notin C$, the set of elements of C incomparable with x is an interval: if $c_i \leq c_j \leq c_k$ and $x \parallel c_i, c_k$, then $x \parallel c_j$.

Indeed, if the incomparable interval of an outside point x contains $r \geq 1$ elements of C , then x contributes r to the left-hand side and $1 + (r - 1)$ to the right-hand side of

$$\sum_{c \in C} |\text{Inc}(c)| = N_C + \sum_{i=1}^{a-1} |\text{Inc}(c_i) \cap \text{Inc}(c_{i+1})|, \quad (4.3)$$

where N_C is the number of points outside C incomparable with at least one point of C . Here $N_C \leq n - a$, while every adjacent intersection in (4.3) has size at most two by CI2. Therefore

$$\sum_{c \in C} |\text{Inc}(c)| \leq n + a - 2.$$

Summing over the three chains counts every incomparable pair exactly twice. If $|C_1| + |C_2| + |C_3| = n$, then

$$2A_2 \leq (n + |C_1| - 2) + (n + |C_2| - 2) + (n + |C_3| - 2) = 4n - 6,$$

which proves the claim. □

Corollary 4.3 (Linear antichain growth). *If P has width three and satisfies CI2, then*

$$|\mathcal{A}(P)| \leq 3n - 2 + \left\lfloor \frac{n - 1}{2} \right\rfloor. \quad (4.4)$$

In particular, the number of antichains is linear in n .

Proof. Width three means $A_3 > 0$. Substitute the bounds of Lemmas 4.1 and 4.2 into $|\mathcal{A}(P)| = 1 + n + A_2 + A_3$. □

Remark 4.4. The linear behavior is a genuine restriction. The disjoint union of three chains of balanced lengths has width three and

$$(|C_1| + 1)(|C_2| + 1)(|C_3| + 1) = \Theta(n^3)$$

antichains. Thus CI2 collapses the possible antichain count from cubic to linear growth within width three.

For $n \leq 14$, Corollary 4.3 already gives a bound strictly below 48. The only remaining issue is the endpoint $n = 15$, where the generic linear estimate gives 50. We need one small tradeoff.

Lemma 4.5 (The fifteen-point refinement). *Let P satisfy the hypotheses above with $n = 15$. If $A_3 \geq 6$, then*

$$|\mathcal{A}(P)| \leq 46.$$

Proof. Order the three-element antichains $T_1, \dots, T_t$ as in Lemma 4.1, put

$$U_i = T_1 \cup \dots \cup T_i, \quad U = U_t,$$

and let m be the number of indices $i < t$ for which $T_{i+1} \cap U_i = \emptyset$. By no re-entry, $T_{i+1} \cap U_i \subseteq T_i$, and distinct triples meet in at most one point. Therefore

$$|U| = 2t + 1 + m. \tag{4.5}$$

Let $e(U)$ denote the number of incomparable pairs with both endpoints in U . We separate two localization steps. First localize the possible new cross edges at a transition $T_i \rightarrow T_{i+1}$. If

$$x \in U_i \setminus T_i, \quad y \in T_{i+1} \setminus U_i,$$

choose $j < i$ with $x \in T_j$. Since $\downarrow T_j \subseteq \downarrow T_i$ and $x \notin T_i$, some member of T_i lies strictly above x ; Lemma 4.1 therefore places x in $L(T_i)$. Analogously, $y \in T_{i+1} \setminus U_i$ cannot lie below a member z of T_i : the ideal inclusion would give $z \leq w$ for some $w \in T_{i+1}$, contradicting that T_{i+1} is an antichain. Thus Lemma 4.1 places y on the upper side of T_i , so $y \in U(T_i)$. By (4.2), $x < y$. Thus every new incomparability joining a new vertex to the old union has its old endpoint in T_i . In what follows, a cross pair excludes pairs already internal to the new triple; in particular a shared vertex is not an old-only endpoint.

If T_i and T_{i+1} meet in one point s , let $x \in T_i \setminus \{s\}$ and $y \in T_{i+1} \setminus \{s\}$. The point x lies in $L(T_{i+1})$ and is incomparable with s , so it must lie below both other members of T_{i+1} ; in particular $x < y$. Hence a nonempty transition creates no cross incomparability.

If the two consecutive triples are disjoint, their bipartite incomparability graph is a matching: each point outside a triple is incomparable with at most one member of that triple. Hence an empty transition creates at most three cross incomparabilities. The new triple itself contributes three incomparable pairs in either case. The accounting is therefore

$ T_i \cap T_{i+1} $	new internal pairs	new cross pairs
1	3	0
0	3	≤ 3

Induction over the transitions gives

$$e(U) \leq 3t + 3m. \tag{4.6}$$

For the second localization step, let $x \notin U$. For every triple T_i , the first paragraph of Lemma 4.1 puts x either in $L(T_i)$ or in $U(T_i)$. These alternatives are monotone along the ideal order: x cannot be below an earlier triple S and above a later triple T . Indeed choose $s \in S$ with $x < s$. Since $\downarrow S \subseteq \downarrow T$, choose $u \in T$ with $s \leq u$. If x were above at least two members of T , one of them, say v , would differ from u , and $v < x < s \leq u$ would contradict the antichain property of T .

Thus the triples above which x lies form an initial segment and the triples below which x lies form a final segment. If both are nonempty, let T_c be the last former triple and T_{c+1} the first latter triple. Every vertex in an earlier triple but outside T_c lies below T_c and therefore below x by (4.2);

dually every later vertex outside T_{c+1} lies above x . Hence all incomparable neighbors of x in U lie in $T_c \cup T_{c+1}$. The first paragraph of Lemma 4.1 gives at most one such neighbor in each triple, so

$$|\text{Inc}(x) \cap U| \leq 2. \quad (4.7)$$

If all triples lie on one side of x , the same argument gives the stronger bound one.

Put $q = 15 - |U|$. Combining (4.6) and (4.7) with (4.5) and the trivial bound on pairs among the q outside points gives

$$A_2 \leq 3t + 3m + 2q + \binom{q}{2}, \quad q = 14 - 2t - m. \quad (4.8)$$

Lemma 4.1 gives $t \leq 7$. If $t = 7$, then $m = q = 0$ and $A_2 \leq 21$. If $t = 6$, then $m + q = 2$; the possibilities $(m, q) = (0, 2), (1, 1), (2, 0)$ give respectively $A_2 \leq 23, 23, 24$. In all cases $t \geq 6$,

$$A_2 + A_3 \leq 30,$$

and (4.1) yields $|\mathcal{A}(P)| \leq 46$. □

We also need the elementary width-two endpoint.

Lemma 4.6 (Width two). *If P has width at most two, satisfies CI2, and $n \geq 2$, then*

$$|\mathcal{A}(P)| \leq 2n + \left\lfloor \frac{n}{2} \right\rfloor - 1.$$

In particular, $|\mathcal{A}(P)| \leq 36$ when $n \leq 15$.

Proof. Partition P into two nonempty chains C, D of sizes $a \leq b$, $a + b = n$. (If P is itself a chain, split it into two nonempty subchains.) As in (4.3),

$$A_2 = \sum_{c \in C} |\text{Inc}(c)| \leq b + 2(a - 1).$$

Since $a \leq \lfloor n/2 \rfloor$,

$$|\mathcal{A}(P)| = 1 + n + A_2 \leq 2n + a - 1 \leq 2n + \left\lfloor \frac{n}{2} \right\rfloor - 1. \quad \square$$

Remark 4.7. The endpoint $n = 15$ is the only place where the separate estimates on A_2 and A_3 are insufficient. Lemma 4.5 uses the ordered family of three-element antichains to couple the two counts; no classification of fifteen-element posets is involved.

Theorem 4.8 (The finite-poset theorem). *Let P be a finite poset satisfying*

$$|P| \leq 15, \quad \text{width}(P) \leq 3, \quad \text{CI2}.$$

Then

$$|\mathcal{A}(P)| \leq 48.$$

Proof. The cases $n \leq 1$ are immediate. If $A_3 = 0$, the width is at most two and Lemma 4.6 applies.

Assume $A_3 > 0$. If $n \leq 14$, then Corollary 4.3 gives $|\mathcal{A}(P)| \leq 46$. Now let $n = 15$. If $A_3 \geq 6$, use Lemma 4.5. If $1 \leq A_3 \leq 5$, Lemma 4.2 gives $A_2 \leq 27$, and hence

$$|\mathcal{A}(P)| = 16 + A_2 + A_3 \leq 16 + 27 + 5 = 48. \quad \square$$

Proposition 4.9 (An explicit extremal poset). *The constant 48 in Theorem 4.8 is attained by $P_* = \{1, \dots, 5\} \times \{1, 2, 3\}$ with strict order*

$$(i, a) < (j, b) \iff j \geq i + 2 \text{ or } (j = i + 1 \text{ and } a \neq b). \quad (4.9)$$

Proof. The relation is irreflexive. In two successive strict comparisons the level increases by at least two in total, which proves transitivity. Thus adjoining equality gives a partial order. Within each level all three points are incomparable. Between consecutive levels exactly the three same-coordinate pairs are incomparable; points at more distant levels are comparable. A mixed-level three-element antichain is impossible: the singleton in one level would have to share its second coordinate with both points in the other level. It follows that the only three-element antichains are the five levels, and none has more than three elements. Hence

$$A_2 = 5 \binom{3}{2} + 4 \cdot 3 = 27, \quad A_3 = 5, \quad |\mathcal{A}(P_*)| = 1 + 15 + 27 + 5 = 48. \quad (4.10)$$

To check CI2, take $(i, a) < (j, b)$. If $j = i + 1$, then $a \neq b$ and the common incomparability set is $\{(i, b), (i + 1, a)\}$, a two-element chain. If $j = i + 2$, that set is $\{(i + 1, a)\}$ when $a = b$, and empty otherwise. If $j \geq i + 3$, it is empty. This checks every comparable pair. $\square$

5 The order-six maximum

Theorem 5.1. *The maximum number of stable matchings in a strict complete stable-marriage instance with six men and six women is*

$$f(6) = 48.$$

Proof. Let R be the rotation poset of an arbitrary order-six instance. Corollaries 2.5 and 2.6, together with Theorem 3.2, give

$$|R| \leq 15, \quad \text{width}(R) \leq 3, \quad R \text{ satisfies CI2.}$$

By Theorem 4.8, R has at most 48 antichains.

For a finite poset, downsets are in bijection with antichains via maximal elements and generated ideals. By the Irving–Leather rotation theorem, stable matchings of the instance are in bijection with downsets of R . Thus the instance has at most 48 stable matchings.

The order-six construction of Benjamin, Converse, and Krieger already has 48 stable matchings [1]. Hence $f(6) = 48$. $\square$

Corollary 5.2 (Necessary conditions for equality). *If an order-six instance has 48 stable matchings, then its rotation poset R satisfies*

$$|R| = 15, \quad A_2 = 27, \quad A_3 = 5. \quad (5.1)$$

Every rotation has exactly two participating men, every man belongs to exactly five rotations, and every man–woman pair occurs in a stable matching.

Proof. The preceding bounds give at most 46 antichains when $|R| \leq 14$, and at most 36 at width at most two. Thus $|R| = 15$ and $A_3 > 0$. Lemma 4.5 excludes $A_3 \geq 6$ at equality. Since $A_2 \leq 27$ and $A_3 \leq 5$, the identity $48 = 16 + A_2 + A_3$ forces $A_2 = 27$ and $A_3 = 5$.

Apply the participation bound to a complete elimination sequence. For each of the six men let d_m be his total number of participations, with $d_m = 0$ for any man in no rotation. Then $d_m \leq 5$, and

$$30 = 2|R| \leq \sum_{\rho \in R} |\text{supp}(\rho)| = \sum_m d_m \leq 6 \cdot 5 = 30. \quad (5.2)$$

Every inequality is an equality. Consequently each support has size two and each d_m equals five. A man sees his initial partner and five subsequent, strictly worse partners along the sequence. These six distinct partners exhaust the six women, and every intermediate matching is stable. Thus all 36 man–woman pairs are attainable. The conclusion is a set of necessary conditions, not a uniqueness or sufficiency assertion. $\square$

Remark 5.3 (The classical example and P_*). The Benjamin–Converse–Krieger DS_3 instance also realizes the abstract sharp example of Proposition 4.9. A direct finite reconstruction of its stable-matching lattice gives 15 join-irreducibles whose poset is isomorphic to P_* ; an explicit isomorphism and reproducible certificate are given in the Supplementary Material. This observation is not used in the upper-bound proof and is not asserted as a uniqueness/classification statement.

6 Verification note

The new finite combinatorial core has accompanying Lean verification. The three formal layers cover the fixed-token participation and small-block bounds, the support-poset implication from explicit capacity hypotheses to width at most three and CI2, and the finite-poset bound of Theorem 4.8, including its fifteen-element endpoint. The formalization is a verification aid, not a logical dependency of the human proof. The connection from actual rotations to the token model and the standard stable-matching–downset correspondence remain human mathematics supported by the classical references above. The Supplementary Material records the precise formalization scope and reproducibility details.

Research provenance and responsibility

Generative-AI systems from OpenAI were used substantively during the research workflow, including exploration of proof strategies and auxiliary lemmas, assistance with finite-verification and Lean-formalization workflows, and drafting or revising parts of the exposition. The human author selected the research objectives and final proof route, checked the mathematical arguments and computational certificates, reviewed the formalization boundaries, and takes full responsibility for every statement in the manuscript. No AI system is an author. The accompanying repository records verification artifacts and the scope of the formalized components. The retained project records do not support a complete exact model/version history, so no such history is asserted here.

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